DNA Extraction from Trichuris trichiura Worm Samples

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Abstract

This protocol outlines the extraction of genomic DNA from *Trichuris trichiura* worm samples for downstream molecular analyses. The workflow includes removal of residual preservatives, mechanical homogenization using bead-based disruption, and DNA extraction optimized for nematode tissues. The resulting DNA is suitable for amplification of the full-length β-tubulin gene and the ITS2 region, supporting applications such as sequencing, phylogenetic analysis, and population genetics.

Guidelines

- Ensure thorough washing of samples to remove residual fixatives (formaldehyde/ethanol), as these may inhibit downstream enzymatic reactions.
- Avoid repeated freeze–thaw cycles of samples prior to extraction.
- Use appropriate PPE when handling biological specimens and chemical preservatives.
- Homogenization efficiency is critical due to the thick nematode cuticle—insufficient disruption may reduce DNA yield.

Materials

Reagents

- Phosphate-buffered saline (PBS), sterile
- Clear-S Quick DNA Extraction Kit (inVIRUStech)
- Nuclease-free water
- 70% ethanol (optional, for cleaning)

Equipements

- TissueLyser LT (QIAGEN) or equivalent bead homogenizer
- Stainless steel beads (3–5 mm)
- Microcentrifuge tubes (1.5 mL or 2.0 mL)
- Pipettes and sterile tips
- Microcentrifuge
- Nanodrop spectrophotometer (or equivalent)
- Liquid nitrogen storage system

Troubleshooting

Problem
Low DNA yield
Solution
Increase cycles or bead size

Problem
Poor PCR amplification
Solution
Repeat washing step

Problem
Low purity ratio
Solution
Extend lysis or improve purification

Problem
DNA degradation
Solution
Maintain cold chain

Safety warnings

- **Risk of PCR inhibition:** Residual preservatives (e.g., formaldehyde or ethanol) from sample storage can inhibit amplification. Ensure thorough washing during DNA extraction.
- **Long-range amplification sensitivity:** The β-tubulin target (~4.2 kb) is sensitive to DNA degradation. Fragmented DNA may fail to amplify even when shorter targets (e.g., ITS2) amplify successfully.
- **Contamination risk:** PCR is highly sensitive. Perform reaction setup in a clean area using sterile filter tips and include a no-template control (NTC) in every run.
- **Non-specific amplification and artifacts:** Excessive cycle numbers or suboptimal primer handling may result in non-specific bands or primer-dimers.
- **Barcoding cross-contamination:** The ITS2 reverse primer contains sample-specific barcodes. Mislabeling or cross-contamination can lead to incorrect sample assignment during downstream sequencing.
- **Polymerase-specific conditions:** This protocol uses a two-step cycling condition optimized for KOD One PCR Master Mix. Using a different polymerase may require re-optimization of annealing and extension conditions.
- **Gel electrophoresis safety:** Handle DNA stains and UV illumination equipment with appropriate precautions to avoid chemical exposure and UV damage.

Samples

1 Sample Adult *Trichuris trichiura* worms preserved in formaldehyde or ethanol during transport

Procedure

2 **Step 1: Sample Washing and Preparation**

2.1 ▪ Transfer worm samples into a clean microcentrifuge tube.

2.2 ▪ Add sterile PBS to fully immerse the sample.

2.3 ▪ Gently invert or pipette to wash.

2.4 Remove PBS and repeat washing 2–3 times to remove residual formaldehyde or ethanol.

3 **Step 2: Sample Storage (if not processed immediately)**

3.1 After washing, snap-freeze samples in liquid nitrogen.

4 **Step 3: Tissue Selection**

4.1 For intact worms:
▪ Excise and use the anterior portion to ensure consistency.

4.2 For fragmented samples:
▪ Use all available tissue material.

5 **Step 4: Mechanical Homogenization**

5.1 Place the sample into a tube containing stainless steel beads (2-4 small beads)

5.2 Add appropriate lysis buffer from the Clear-S kit.

5.3 Homogenize using TissueLyser LT:
- Frequency: **50 Hz**
- Duration: **1 minute per cycle**
- Number of cycles: **2–3 cycles**

5.4 Briefly centrifuge between cycles if needed.

6 **Step 5: DNA Extraction**

6.1 Proceed with DNA extraction using the Clear-S Quick DNA Extraction Kit following manufacturer's instructions.

6.2 Apply an extended lysis step to improve DNA recovery:
- Extend incubation time as needed (e.g., +10–20 minutes beyond standard protocol).

7 **Step 6: DNA Elution**

7.1 Elute DNA in nuclease-free water or provided elution buffer.

7.2 Typical elution volume: 30–50 µL.

8 **Step 7: DNA Quantification and Quality Assessment**

8.1 Measure DNA concentration using a Nanodrop spectrophotometer.

8.2 Record:
- DNA concentration (ng/µL)
- Purity ratios (A260/A280)

9 **Step 8: Storage of Extracted DNA**

9.1 Store extracted DNA at:
- −20 °C (short-term)
- −80 °C (long-term)

Expected Results

10 High-quality genomic DNA suitable for:
- Long-range PCR (β-tubulin ~4.2 kb)
- ITS2 amplification (~481 bp)

10.1 DNA concentration sufficient for downstream applications (>10–20 ng/µL recommended)